

Context Diagram — Heart Failure Remote Patient Monitoring System (HF-RPM)

What Is a Context Diagram?

A context diagram is the highest-level model of a system. It represents the system of interest (Sol) as a single undifferentiated *black box*, draws an explicit boundary around it, and shows every external entity the system exchanges something with, together with the labeled, directional flows that cross that boundary. Its defining characteristic is what it deliberately withholds: it says nothing whatsoever about internal components, functions, or architecture. INCOSE defines the system boundary as the “line of demarcation” between the system under consideration and its greater context, and states that the boundary determines what belongs to the system and what does not. The context diagram is the artifact that makes that line explicit, visible, and reviewable by people who would otherwise each assume a different one.

Its notation is intentionally minimal — one box for the Sol, one box per external entity, one labeled arrow per flow. That economy is the point. Because the diagram cannot express internal design, it cannot be used to argue about internal design, which forces a team to finish the question of scope before opening the question of solution.

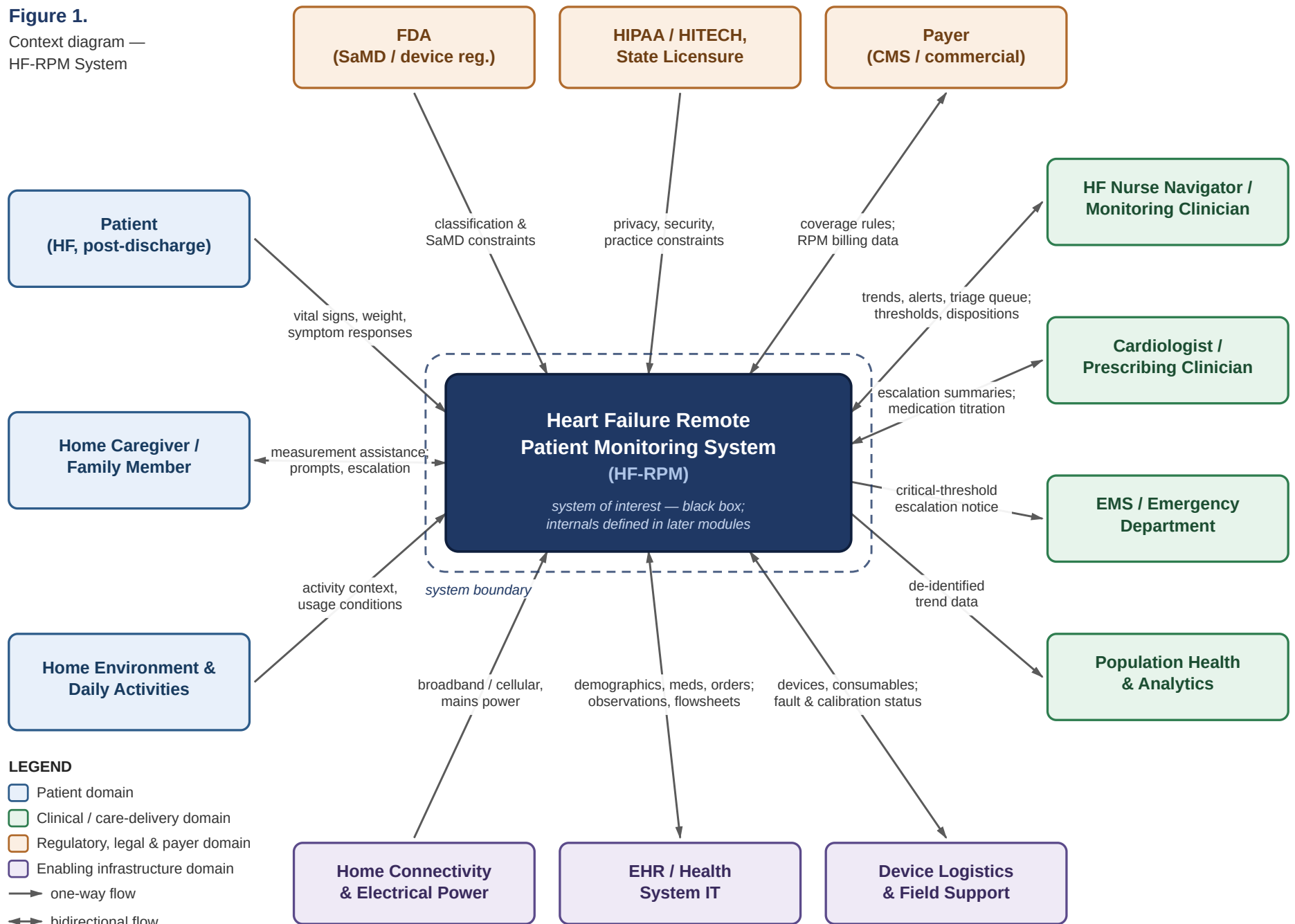
What Is Its Purpose?

1. **Fix scope and boundary before design begins.** Every element is either inside the box or outside it, and the diagram makes that commitment visible. This is the earliest and cheapest defense against both scope creep and its quieter twin, scope gaps — the interfaces nobody claimed because nobody drew them.
2. **Enumerate the full stakeholder set.** Building the diagram forces the question “who or what else touches this?” until the answers run out. It routinely surfaces entities no single stakeholder would have volunteered — the payer, device logistics, EMS — which is a structural remedy for the blind-men-and-the-elephant problem, where each participant describes only the part of the system they can personally see.
3. **Identify every external interface.** Each arrow crossing the boundary is a future interface requirement. Interfaces are where systems most often fail and where integration cost concentrates, so enumerating them at the beginning is disproportionately valuable relative to the effort involved.
4. **Seed operational needs and requirements.** The flows are the raw material for the needs and requirements developed later in the lifecycle: what the system must accept, what it must produce, at what rate, and under what conditions. The diagram is therefore the first input to the systems engineering method.
5. **Expose constraints that are not negotiable.** Laws, regulations, reimbursement rules, and existing infrastructure are not design choices; they are givens that the design must satisfy. Placing them on the diagram prevents a team from discovering them late, when accommodating them is most expensive.
6. **Create a shared mental model.** A single page a clinician, an engineer, a compliance officer, and an executive can all read and correct is how a systems view becomes shared rather than privately held.

Selected Remote Healthcare System

The system of interest is a **heart failure remote patient monitoring system (HF-RPM)**: a home-based capability that monitors recently discharged heart failure patients for early signs of decompensation — principally weight gain from fluid retention, together with blood pressure, heart rate, oxygen saturation, and structured symptom reporting — and escalates to a clinical team in time for outpatient intervention rather than readmission. Heart failure patients carry roughly a 40% readmission rate at 90 days, and the physiologic signal that precedes decompensation typically appears days before the patient presents. The gap is not diagnostic knowledge but a sensing-and-escalation capability operating in the home. The environment spans the tiers introduced in this module: primarily home-based care, with a clinical monitoring node and, at the escalation limit, emergency response.

Figure 1.
Context diagram —
HF-RPM System



Every arrow crossing the boundary is a candidate interface requirement.

Interface Summary — Flows Crossing the System Boundary

External Entity	Domain	Dir.	Content of the flow
Patient (HF, post-discharge)	Patient	In	Daily weight, blood pressure, heart rate, SpO ₂ , and structured symptom-survey responses (dyspnea, orthopnea, edema).
Home Caregiver / Family Member	Patient	Bi	In: assistance with measurement, confirmation of readings. Out: adherence prompts, plain-language guidance, escalation notification.
Home Environment & Daily Activities	Patient	In	Ambient conditions, measurement timing and routine, competing devices, physical layout and network placement.
HF Nurse Navigator / Monitoring Clinician	Clinical	Bi	Out: trended physiologic data, threshold and trajectory alerts, prioritized triage queue, adherence status. In: patient-specific thresholds, care-plan updates, alert dispositions.
Cardiologist / Prescribing Clinician	Clinical	Bi	Out: escalation summaries and longitudinal trend reports. In: medication titration decisions, revised monitoring intent.
EMS / Emergency Department	Clinical	Out	Critical-threshold escalation notice with recent trend context, at the point where remote management is no longer sufficient.
Population Health & Analytics	Clinical	Out	De-identified longitudinal trend data supporting cohort analysis, program evaluation, and algorithm performance monitoring.
FDA (SaMD / device regulation)	Governance	In	Device classification, software-as-a-medical-device constraints, clinical-decision-support boundaries, labeling and change-control obligations.
HIPAA / HITECH, State Licensure	Governance	In	PHI privacy and security obligations, audit-logging requirements, breach-notification duties, cross-state telehealth practice constraints.
Payer (CMS / commercial)	Governance	Bi	In: coverage criteria and RPM billing rules (e.g., minimum transmission days, interactive-communication time). Out: billing-qualifying data and time documentation.
Home Connectivity & Electrical Power	Infrastructure	In	Broadband or cellular service of variable reliability; mains power subject to outage. Both are given conditions, not design choices.
EHR / Health System IT	Infrastructure	Bi	In: patient demographics, problem list, medication list, discharge orders, identity and access control. Out: observations, flowsheet entries, alerts, documentation.
Device Logistics & Field Support	Infrastructure	Bi	Out: device fault status, battery and calibration state, return triggers. In: enrolled kits, consumables, replacements, on-site or remote technical support.

Dir. — In: entity to system. Out: system to entity. Bi: bidirectional.

Boundary Decisions — What Is Deliberately Outside the Box

A boundary is only meaningful if the exclusions are stated as explicitly as the inclusions. The following are outside the HF-RPM boundary by decision, not by oversight.

Excluded from the Sol	Rationale
The patient's implanted or wearable cardiac devices	Separate regulated systems with their own manufacturers and data pathways; HF-RPM may receive their data but does not control or maintain them.
The EHR itself	HF-RPM exchanges data with it across a defined interface. Absorbing the EHR into the boundary would make the project unbounded.
Inpatient care and the index hospitalization	HF-RPM begins at discharge. The hospitalization is the triggering event, not part of the system.
Clinical decision-making authority	The system surfaces information and escalates; licensed clinicians decide. This boundary is a regulatory constraint, not a preference.
Payer adjudication and claims processing	HF-RPM supplies documentation; adjudication happens outside the boundary.

Assumptions

- The deployment setting is home-based care in the United States, with a clinical monitoring node operated by the discharging health system.
- Enrollment is post-discharge and time-bounded (on the order of 30–90 days), not lifelong monitoring.

- The system is informational and escalatory; it does not deliver therapy and does not exercise autonomous clinical decision authority.
- Patients may have limited technical proficiency, intermittent connectivity, and variable caregiver support. These are context conditions the design must tolerate, not assumptions to be engineered away.

References

INCOSE. (2023). *Systems engineering handbook* (5th ed.). INCOSE-TP-2003-002-05. · Kossiakoff, A., Biemer, S. M., Seymour, S. J., & Flanagan, D. A. (2020). *Systems engineering principles and practice* (3rd ed.). Wiley. · Montoya, M. Module 1: Framework, context, and motivation [Course lecture]. Johns Hopkins University. · Pronovost, P. (2019). *New narratives for health care* [Address]. The City Club of Cleveland.